**Lebanese American University**




COE547 Deep Learning

# Hypoglycemia Detection from ECG Signals using Deep Learning

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# 1 Introduction

Hypoglycemia is a critical and potentially dangerous condition caused by abnormally low blood glucose levels. Detecting hypoglycemia early is essential to prevent severe neurological complications. Recent research has shown that physiological signals, especially ECG, contain measurable patterns that correlate with glucose fluctuations.

In this work, we analyze a dataset of ECG recordings collected from nine different patients, each with varying glucose conditions across multiple days. The task is formulated as a binary classification problem: detecting whether a 10-second ECG segment corresponds to a hypoglycemic event.

The dataset is highly imbalanced both across patients and within each patient, motivating the need for advanced preprocessing, normalization, and synthetic data generation. We adopt a 1D CNN + BiLSTM classifier combined with GAN-based augmentation to address class imbalance and improve generalization.

# 2 Dataset Description

Each sample in the dataset corresponds to a 10-second ECG segment containing 2500 time-domain samples. Additional metadata includes:

- `patient_id`
- `day`
- `count`
- `glucose_values`
- `label` (0 = normal, 1 = hypoglycemia)

Table 1 summarizes the extreme imbalance present in the dataset.

Table 1: ECG Sample and Hypoglycemic Event Distribution per Patient

| **Patient ID** | **Total Samples** | **Hypo Samples** |
|---|---|---|
| 1 | 569 | 54 |
| 2 | 4398 | 372 |
| 3 | 1723 | 0 |
| 4 | 2326 | 358 |
| 5 | 44 | 0 |
| 6 | 5621 | 0 |
| 7 | 14363 | 3 |
| 8 | 16560 | 1674 |
| 9 | 26 | 0 |

The imbalance is severe: some patients have no positive samples at all, while patient 8 alone accounts for more than 1600 hypoglycemic segments.

# 3 Methods

This covers preprocessing, feature scaling, stratified splitting, augmentation, model structure, and threshold tuning.

## 3.1 Stratified Splitting and Removal of Patient Identifiers

We used a stratified train–validation split to keep the overall ratio of hypoglycemic and non-hypoglycemic samples the same in both sets. This makes the validation results (F1-score, recall, precision) more reliable.

We also removed the `patient_id` column before training. This was not mainly for data-leakage reasons, but because the dataset is very uneven: some patients have tens of thousands of ECG segments, while others have only a few. If the model could see `patient_id`, it would focus on the large-patient groups and ignore the smaller ones, leading to a biased model that does not generalize.

By removing the identifier, the model is forced to learn patterns from the ECG signals themselves instead of memorizing patient identity. We tried stricter methods like patient-wise splits and leave-one-subject-out, but because positive events are so rare, these splits produced models that were unstable or unable to detect hypoglycemia at all.

In highly imbalanced medical datasets, it is sometimes acceptable to relax strict patient separation during early experiments. In our case, merging all patients and removing the identifier allowed the model to finally learn useful ECG patterns related to hypoglycemia.

## 3.2 Normalization

Each ECG segment is normalized independently using:

$$x' = \frac{x - \mu}{\sigma}$$

This reduces amplitude bias between patients and recording sessions.

## 3.3 GAN-Based Positive Augmentation

Given the extreme imbalance (30218 negatives vs 1723 positives), a Generative Adversarial NEtwork (GAN) was trained to generate synthetic hypoglycemic ECG segments. GANs are a deep-learning framework widely used in deep learning approaches in which a generator and discriminator are trained adversarially to learn data distributions and synthesize realistic samples [1]. The generator takes a random latent vector and progressively upsamples it through dense layers, reshaping, 1-D convolutions, and LeakyReLU activations over ReLU to avoid dead neurons until it produces a 2500-sample ECG-shaped signal. The discriminator is a 1-D convolutional classifier that receives either a real positive ECG window or a generator-produced one and outputs a probability of being real. Both models are optimized: the discriminator is trained with binary cross-entropy to distinguish real from fake, while the generator is trained through a combined model where the discriminator is frozen, forcing the generator to adjust its weights to make its outputs look increasingly realistic to the discriminator [1]. Over many iterations of this two-step training loop, the generator learns to synthesize ECG segments whose structure matches the real hypoglycemic examples it was trained on. GAN evaluation metrics:

Table 2: GAN realism evaluation

| Metric | Mean | Std |
|---|---|---|
| Real Score | 0.9083 | 0.0407 |
| Fake Score | 0.1119 | 0.0587 |

As the above table shows, the real score is high and the fake score is low indicating that the discriminator was effectively distinguishing between fake and real samples, which as a result means it was feeding the generator correct updates. After augmentation, positives increased from 1723 to approximately 5169.

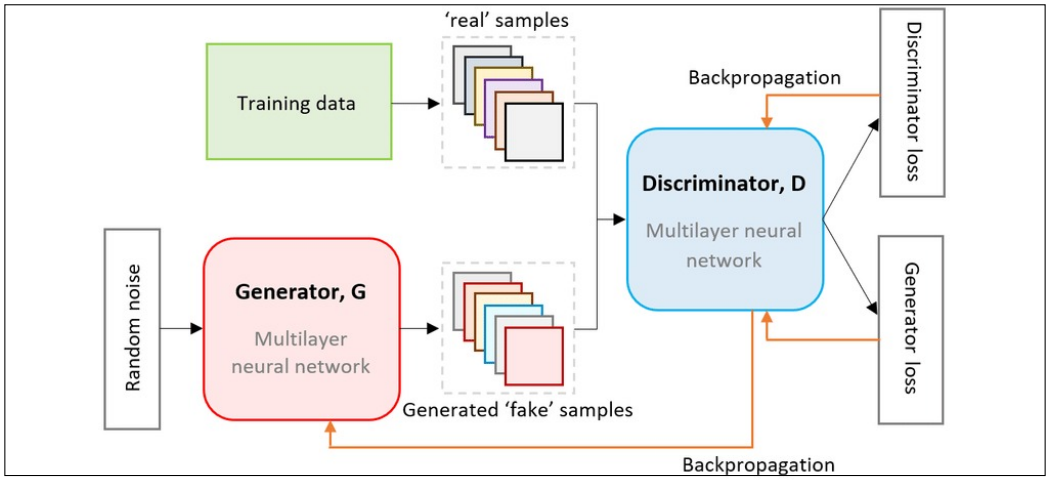



Figure 1: GAN model structure showing the generator–discriminator setup.

## 3.4 Final Train/Validation Split After Augmentation

Table 3: Dataset size after GAN augmentation

| Set | Total Samples | Positives | Positive Ratio |
|---|---|---|---|
| Training | 35387 | 5169 | 0.146 |
| Validation | 8847 | — | — |

## 3.5 Model Architecture Summary

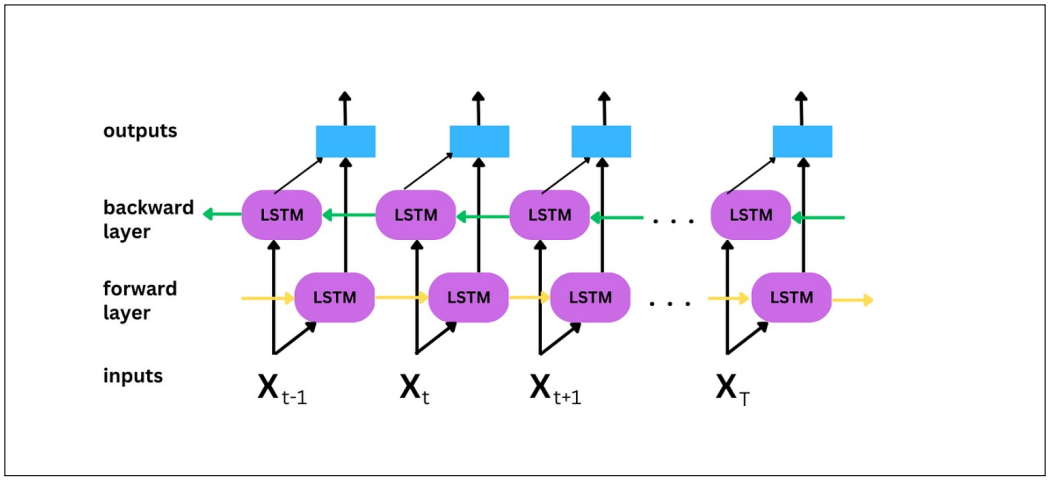



Figure 2: BiLSTM network structure showing forward and backward recurrent flows.

A Bidirectional Long Short-Term Memory (BiLSTM) network is a recurrent neural architecture that processes sequence data in both forward and backward directions. Unlike standard LSTMs, which capture only past dependencies, BiLSTMs learn relationships from both past and future time steps, making them particularly effective for physiological signals such as ECGs where patterns depend on context across the entire waveform. By combining information from both directions, the BiLSTM improves the model's ability to detect subtle temporal features relevant to classification tasks. [2]

| Layer (type) | Output Shape | Parameters |
|---|---|---|
| InputLayer | (None, 2500, 1) | 0 |
| Conv1D (32 filters, kernel=7) | (None, 1250, 32) | 256 |
| BatchNormalization | (None, 1250, 32) | 128 |
| ReLU | (None, 1250, 32) | 0 |
| MaxPooling1D | (None, 625, 32) | 0 |
| Conv1D (64 filters, kernel=7) | (None, 625, 64) | 14,400 |
| BatchNormalization | (None, 625, 64) | 256 |
| ReLU | (None, 625, 64) | 0 |
| MaxPooling1D | (None, 312, 64) | 0 |
| Conv1D (128 filters, kernel=7) | (None, 312, 128) | 57,472 |
| BatchNormalization | (None, 312, 128) | 512 |
| ReLU | (None, 312, 128) | 0 |
| MaxPooling1D | (None, 156, 128) | 0 |
| Bidirectional LSTM (64 units) | (None, 156, 128) | 98,816 |
| GlobalAveragePooling1D | (None, 128) | 0 |
| Dense (64 units) | (None, 64) | 8,256 |
| Dropout (rate=0.3) | (None, 64) | 0 |
| Dense (1 unit, sigmoid) | (None, 1) | 65 |
| **Total Parameters** | – | **180,161** |

Table 4: Final 1D-CNN + BiLSTM Architecture Used for ECG-Based Hypoglycemia Detection.

### 3.6 Threshold Optimization Algorithm

We sweep thresholds from 0.05 to 0.95 and choose the one maximizing F1 by testing over the validation set and choosing the threhold with the best f1. We found that the best threshold turned out to be 0.85, which we used for the testing part.

## 4 Experimental Results

The results are as follows:

Table 5: Test Metrics (Random Split, No Patient-Wise Separation)

| Metric | Value |
|---|---|
| Accuracy | 0.977 |
| Recall (Hypo) | 0.759 |
| Precision (Hypo) | 0.805 |
| F1 (Hypo) | 0.781 |
| AUROC | 0.981 |

Table 6: Confusion matrix of the CNN–BiLSTM model under random stratified splitting. Rows represent true labels and columns represent predicted labels.

| **True / Pred** | **Pred = 0** | **Pred = 1** |
|---|---|---|
| **True = 0** | 6408 | 68 |
| **True = 1** | 89 | 280 |

# 5 Discussion

The results show both the promise and the difficulty of detecting hypoglycemia using ECG signals. The main challenge comes from the dataset itself: hypoglycemic events are very rare, and some patients contribute thousands of ECG segments while others contribute almost none. Without synthetic data, the model mostly learns to predict the majority class and therefore misses many positive cases. Using a GAN to generate extra positive samples helped balance the training data and allowed the model to learn clearer ECG patterns linked to hypoglycemia. This led to better recall and a more meaningful F1 score.

Removing the `patient_id` column also helped. Because the patients are so unevenly represented, keeping the identifier would cause the model to rely on patient identity instead of ECG morphology. Treating all samples as one combined dataset gave the model a better chance to learn general ECG patterns rather than overfitting to the largest patient groups. More strict approaches such as patient-wise splitting or leave-one-subject-out were tested, but these methods left the model with too few positive samples to learn from and resulted in very poor performance.

With the final setup, the classifier reached an F1 score of 0.781 and an AUROC of 0.981. These results show that the ECG contains useful information for detecting hypoglycemia even though the dataset is highly imbalanced. We also tried a more advanced generator, TimeGAN, but in this case the simpler GAN produced better results. Overall, the project shows that ECG-based hypoglycemia detection is possible, but performance is still limited by the small number of hypoglycemic events. A more balanced dataset or better synthetic data methods would likely improve results in future work.

# 6 Previous Approaches

Before arriving at the final adopted methodology, several alternative approaches were also attempted, including the following:

## 6.1 Random Stratified Splitting Without Patient Separation

Before adopting the GAN-augmented pipeline, we first evaluated a simpler baseline approach that also used stratified random splitting and removed the `patient_id` column. Both approaches share the same motivation: the extreme imbalance across patients made patient-wise splitting infeasible for stable training, and treating the dataset as a unified pool allowed the model to learn global ECG patterns rather than patient-specific distributions.

However, unlike the final GAN-based method, this baseline used only classical augmentation and relied purely on the available positive samples, without generating syn-

thetic ones. As a result, the class ratio remained limited by the scarce hypoglycemic events, especially for most patients who had very few or zero positives.

Under this setup, a CNN–BiLSTM model was trained with per-sample normalization, stratified splitting, and AUC-based early stopping. Threshold tuning was then performed on the validation set, and the resulting metrics are shown in Table 7. Although the performance is strong, it is still a bit constrained by the natural imbalance, unlike the final GAN-enhanced approach, which benefits from balanced synthetic positives and achieves more stable recall during training.

Table 7: Performance of the CNN–BiLSTM baseline under random stratified splitting (no patient separation, no GAN augmentation).

| **Metric** | **Value** |
|---|---|
| Accuracy | 0.974 |
| Recall (Hypo) | 0.740 |
| Precision (Hypo) | 0.765 |
| F1 Score | 0.752 |
| AUROC | 0.974 |

## 6.2 Usage of TimeGAN instead of GAN

In this version of the experiment, we replaced the previous GAN-based data generation method with a model called TimeGAN. Unlike a standard GAN, which generates new ECG segments directly, TimeGAN creates synthetic data by learning both the shape of the signal and how the signal changes over time. This makes it more suitable for time-series data such as ECG. We made this change to try to produce more realistic synthetic positive samples that could help the classifier learn better from an imbalanced dataset. However, TimeGAN is a more complex model and requires longer training because it first learns to reconstruct signals, then learns temporal patterns, and finally trains the generative and discriminative parts together. Despite the improved structure, the synthetic samples it produced in our setup did not lead to a higher F1 score than the simpler GAN approach. The slightly lower performance suggests that either the TimeGAN training was not fully optimized for our dataset or that the classifier benefited more from the simpler GAN-generated samples in this specific case.

Table 8: Performance with TimeGAN

| **Metric** | **Value** |
|---|---|
| Accuracy | 0.971 |
| Recall (Hypo) | 0.720 |
| Precision (Hypo) | 0.745 |
| F1 Score | 0.732 |
| AUROC | 0.981 |

## 6.3 Other Approches

In addition to the previously mentioned, multiple other approaches were used. We started off with patient wise split using 1d CNN and BiLSTM, but that lead to very poor results.

After that we attempted to use feature extraction, where we extracted the mean, standard deviation, frequency features and many others from the raw 10 second signals and input them into a regular MLP. The best f1 score that achieved was around 60%. After that, we went to the approaches that included dropping aptient ID, using stratify for the data split and using different data augmentation techniques, include SMOTE, GAN, and basic augmentation techniques such as scaling and jittering.

# 7 Conclusion

We built a deep learning pipeline for hypoglycemia detection using 10-second ECG signals, addressing severe class imbalance with GAN-based augmentation and careful preprocessing. The 1D CNN + BiLSTM classifier achieved reasonable F1 and recall despite the challenging dataset. This work demonstrates the feasibility of ECG-only hypoglycemia detection and highlights promising directions for future improvements.

# References

[1] K. Ahirwar, *Generative adversarial networks projects: Build next-generation generative models using TensorFlow and Keras*. Packt Publishing Ltd, 2019.

[2] A. A. Sharfuddin, M. N. Tihami, and M. S. Islam, "A deep recurrent neural network with bilstm model for sentiment classification," in *2018 International Conference on Bangla Speech and Language Processing (ICBSLP)*. IEEE, 2018, confidence and performance metrics reported in the original publication.